

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Theory of Computation (New)

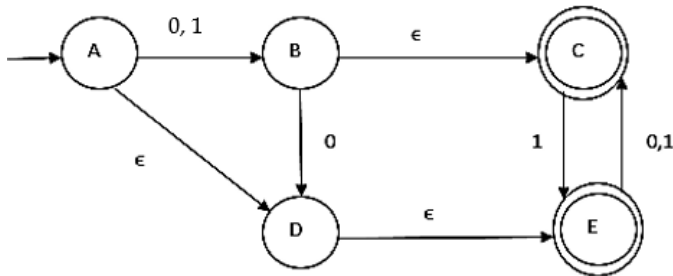
Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) State the Pigeonhole Principle. Provide a formal proof for the following: "In any group of n people, there must be at least two people who have the same number of friends within that group." 7
b) What is a regular expression? Construct a regular expression for the language of all strings over $\{a,b\}$ containing exactly two a 's. Also draw a DFA for the expression. 8
2. a) Explain Finite Automata? Design a DFA that accepts the language given by $L = \{w \in \{a,b\}^* : w \text{ does not contain three consecutive } b\text{'s}\}$. Hence test your design for **abaabb** and **babbbba**. 7
b) Construct a DFA equivalent to NFA as shown: 8



3. a) Define a Context-Free Grammar (CFG). Construct a CFG for the language $L = \{a^n b^m c^n \mid n > 1, m > 1\}$. Provide the formal 4-tuple $\{V, T, P, S\}$ for your grammar. 7
b) Apply the pumping lemma to prove $L = \{a^n b^n c^n \mid n \geq 0\}$ is not context-free. Show all steps of the proof. 8

OR

Explain important properties of context-free languages (CFLs) with examples.

4. a) State decision properties of CFL. Show that CFL are not closed under complementation. 7
- b) Define PDA formally. Design a PDA for $L = \{ a^n b^{2n} : n > 0 \}$. 8
5. a) How can multi-tape turing machine be simulated through single tape turing machine? Explain with an example. 7
- b) Design a Turing machine for the following language: $L = \{ w \in \{a,b\}^* / w \text{ has equal number of } a\text{'s and } b\text{'s} \}$. 8

OR

Design a Turing machine which works as copying machine for $w \in \{a, b\}^+$.

6. a) Define a Universal Turing Machine (UTM). Explain how it simulates other Turing Machines. 7
- b) Define tractable and intractable problems. Explain why finding a polynomial-time solution for any NP-complete problem would solve all NP problems. 8
7. Write short notes on: (**Any two**) 2×5
 - a) Chomsky Hierarchy
 - b) Unrestricted grammar
 - c) Halting Problem
 - d) Complexity Analysis